Automotive Cyber Security Academy 2023

Basic Information Technology Security

Topic 5: Security Standards

Automotive Cyber Security Academy

1 2 9 0 FACULDADE DE CIÊNCIAS E TECNOLOGIA UNIVERSIDADE D COIMBRA

dei departamento de engenharia informática

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Outline

- The role of governments
- Security related standards

The role of governments

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- National Cyber Security Agencies
  - Centro Nacional de Ciber Segurança – CNCS (Portugal)
- CyberSecurity Frameworks
- Legislation
- Enforcement and Protection

National Cybersecurity Framework

- Risk-based approach to tackle cyber threats
- Guidance for decision makers and IT departments
- Security Measures:
  - Identify
  - Protect
  - Detect
  - Respond
  - Recover

National Cybersecurity Framework

- Roles for teams/individuals:
  - CISO – Chief Information Security Officer
  - SOC – Security Operation Centre
  - CSIRT – Cybersecurity Incident Response Team

Security Related Standards and Initiatives

European Security Bodies

- CSIRTs-EU – Computer Security Incident Response Team from public and private entities of the European countries.
- ENISA – Coordinates and synchronizes the operational cooperation and technical between actors in the UE space, Coordinate the Computer Security Incident Response Teams.
- CyCLONe – Established Network for Connection in crises in the European Cyberspace.
- EUROPOL – Police Body organized to provide response to the crimes performed in the cyberspace. Also dedicated with forensics activities.

Security Bodies

- NCIRC – NATO Computer Incident Response Capability, transatlantic alliance for collective defence.
- OSCE – Organization for Security and Co-operation in Europe, activities cover conflict prevention, promotion of full respect of human rights.
- NIST – National Institute of Standards and Technology from US

ENISA Security Documents

- 5G Cybersecurity Standards
- Risk Management Standards
- Security in 5G Specifications – Controls in 3GPP
- Others…

GDPR – General Data Protection Regulation

- Defined in the EU Directive 95/46/EC in 27/April/2016
- Put in Effect on May 2018
- Privacy and security standard
  - Defined in 11 Chapters

GDPR

Chapter 1 (Art. 1 – 4)
General provisions

Chapter 2 (Art. 5 – 11)
Principles

Chapter 3 (Art. 12 – 23)
Rights of the data subject

Chapter 4 (Art. 24 – 43)
Controller and processor

Chapter 5 (Art. 44 – 50)
Transfers of personal data to third countries or international organisations

Chapter 6 (Art. 51 – 59)
Independent supervisory authorities

Chapter 7 (Art. 60 – 76)
Cooperation and consistency

Chapter 8 (Art. 77 – 84)
Remedies, liability and penalties

Chapter 9 (Art. 85 – 91)
Provisions relating to specific processing situations

Chapter 10 (Art. 92 – 93)
Delegated acts and implementing acts

Chapter 11 (Art. 94 – 99)
Final provisions

Reflection

Reflections

- Identify the organism, responsible for security enforcement in your country ?
- Which organizations are working on security ?
- Which standards are more relevant for me ?

Thanks

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The NEW and IMPROVED - ACSA 2023 is here!